Deployment Test Report: Four-Bay Truss

Test campaign DT-07 · Prepared for internal review

1. Summary

The four-bay truss deployed fully in all six ground tests. Deployment took 11.8 s on average, within the 15 s requirement. Peak latch load stayed under 40% of the allowable. One test showed a brief hinge bind at bay 3, cleared by the spring margin without intervention.

2. Deployment sequence

Figure 1 shows the panel stack unfolding from its stowed state. The motion is the same sequence the flight unit will perform after separation.



Figure 1. Panel deployment sequence (animated).

For comparison, the stowed configuration before release:

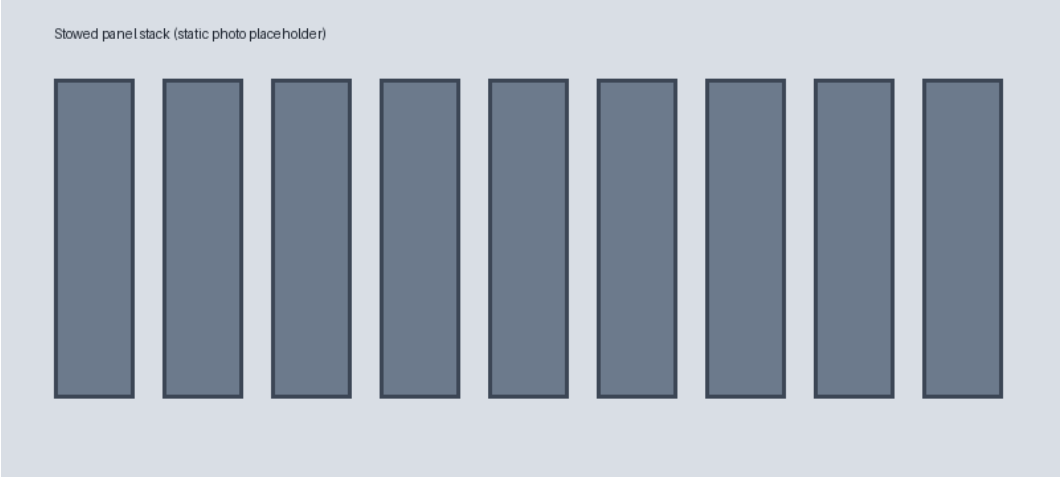


Figure 2. Stowed configuration (still image).

3. Results

Test	Deploy time (s)	Peak latch load (N)	Result
DT-07-1	11.2	182	Pass
DT-07-2	12.1	190	Pass
DT-07-3	13.4	204	Pass, hinge bind
DT-07-4	11.0	176	Pass
DT-07-5	11.6	185	Pass
DT-07-6	11.5	181	Pass

4. As-built geometry

Figure 3 turns the as-built truss through a full revolution so reviewers can check bay alignment from every side without opening CAD.

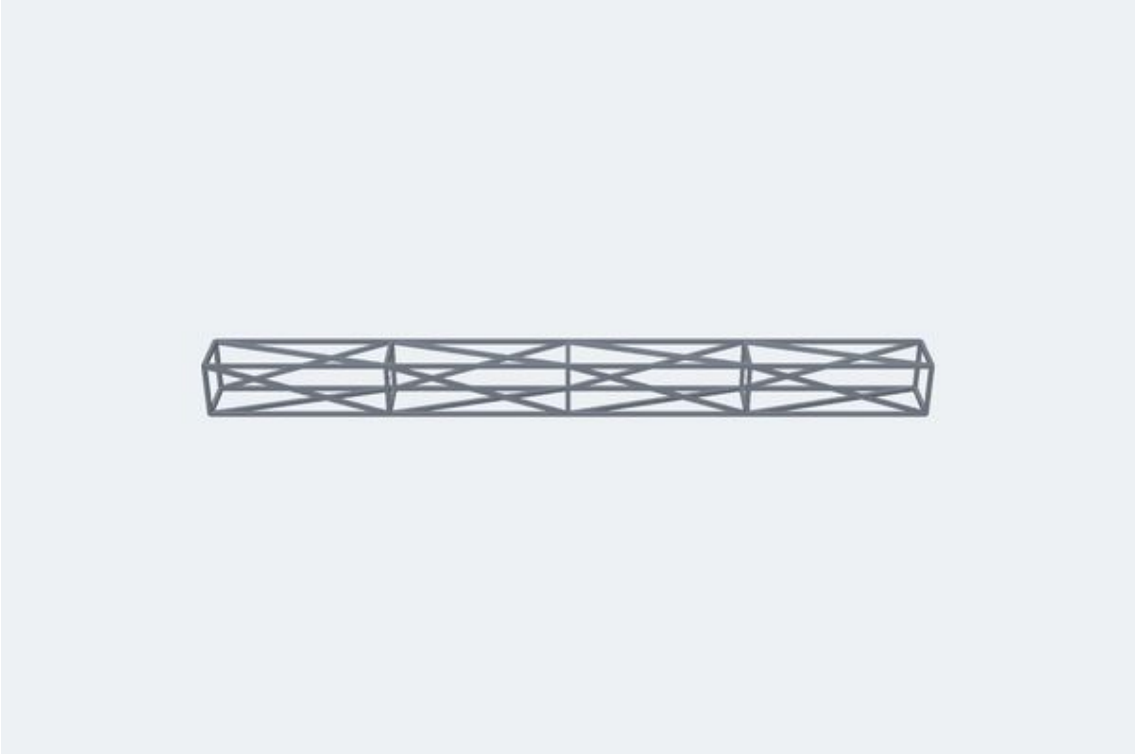


Figure 3. As-built truss, full revolution (animated).

5. Open items

- Root-cause the bay 3 hinge bind (suspected burr on pin, inspect before DT-08).
- Repeat DT-07-3 with the replacement pin.
- Add thermal soak at -40 °C before the next deployment series.

6. Conclusion

The design meets its deployment requirements on the ground. The team recommends proceeding to thermal vacuum testing once the hinge bind is closed out.